

A Stateless, Parameterized Hybrid-Information Retrieval (HIR) System.

Software Requirements Specifications for WP_DataAnswer

1. Intro

1.1 Purpose

The objective of this document is to specify the requirements for WP_DataAnswer, an intelligent data processing system that performs automated differentiation between structured (Excel/CSV) and unstructured (Text) sources. Unlike generic solutions, this system is designed to function as a dynamic intermediary that transforms the user's natural language into precise logical operations or semantic searches. This document serves as the foundation for developing a platform that grants the owner full control over resources without tethering them to a fixed infrastructure.

1.2 Scope

WP_DataAnswer focuses on three core pillars: precise context extraction, user data isolation, and administrative flexibility.

- **Intelligent Routing:** The system independently determines whether a query requires a descriptive answer from a textual document or a mathematical aggregation from a tabular dataset.
- **Dynamic Parameterization:** The operational scope (limits, session duration, workload) is not hardcoded into the software but depends on variables defined by the owner and the capacity of their server.
- **Ephemeral Processing:** The system is limited to session-based operations—its scope does not include long-term document archiving, but rather rapid, "on-the-fly" analysis with total data deletion following every interaction.
- **Adaptive Authentication:** The platform supports user hierarchy, allowing the owner to allocate unlimited resources to specific profiles (Premium) while controlling performance for others.

2. Overall Description

Product Perspective: The system is established as a dynamic platform where every key capacity decision is extracted into external configuration variables. This allows the administrator to easily adjust the system without modifying the source code.

Administrative Control and Domain Knowledge: The owner or the individual managing the system must possess domain knowledge regarding the infrastructure (server power, memory, and capacity). Based on this knowledge, variable values are set to define system limits and ensure stable operation.

User Classes:

- **Standard User:** A user subject to the current system restrictions defined by the administrator.

- Premium User: A user with elevated privileges, where the owner has the option to completely remove all limits (number of files, size, duration).
- Administrator: A person authorized to configure global variables and manage user statuses.

3. System Features

3.1 Parameterized Resource Management

The system relies on variables to dynamically define the following parameters:

- File Capacity: The maximum number of allowed files per user conversation.
- Data Volume: Limits for individual file size and total data volume within a single session.
- User Throughput: The maximum number of users who can simultaneously start a new conversation, depending on server capacity.
- Session Timeframe: A defined inactivity period after which the system flushes data from memory.

3.2 Authentication and Isolation

Access to functionalities is enabled via authentication. Each user session is strictly isolated, allowing for the parallel operation of multiple users without the risk of data cross-contamination.

4. Non-functional Requirements

4.1 Privacy and Data Management

Session Memory: All uploaded data and contexts are temporary. Upon session termination, logout, or expiration of the set time, the system deletes all data associated with that user.

Protection Policy: The system maintains no logs regarding the content of conversations and does not transmit information to external storage systems.

4.2 Performance

Adaptability: Processing time and system responsiveness are directly linked to the owner's hardware resources. The system offers the freedom to adjust these performance metrics according to specific needs.

5. External Interface Requirements

5.1 User Interface (UI)

A clear display of the current capacities allocated to the user.

An Export Chat function to save the conversation before its permanent deletion.

5.2 Configuration

Interface Enables easy modification of system variables by the owner.